

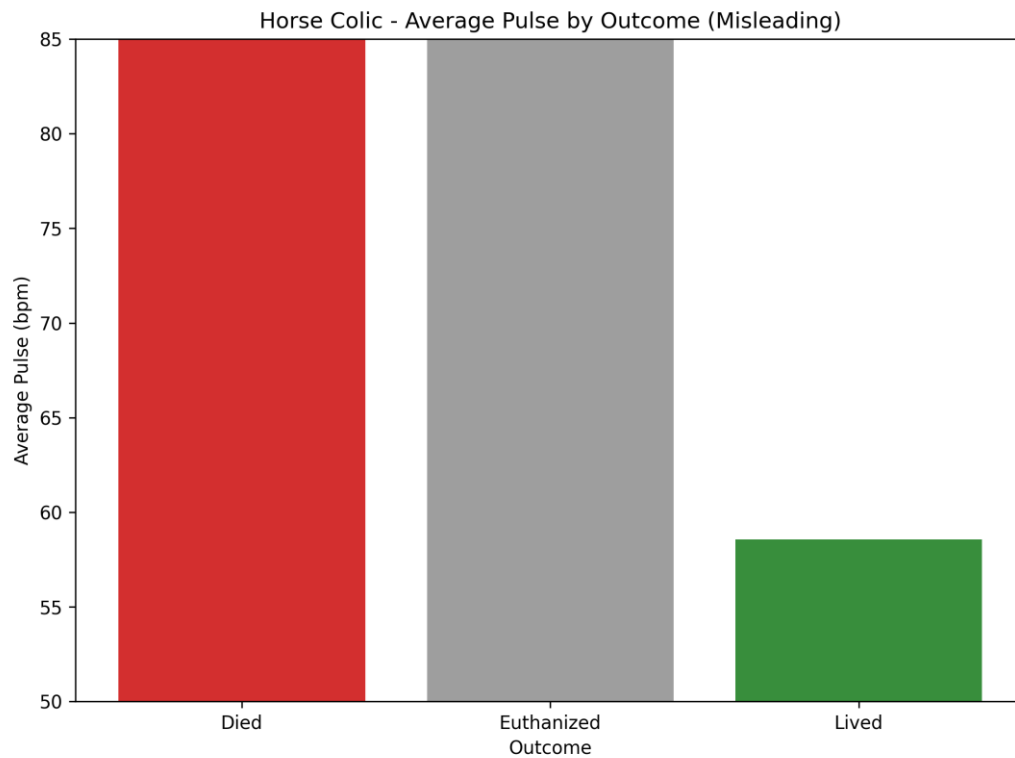


HOMework 5: CORRECTING MISLEADING VISUALIZATIONS

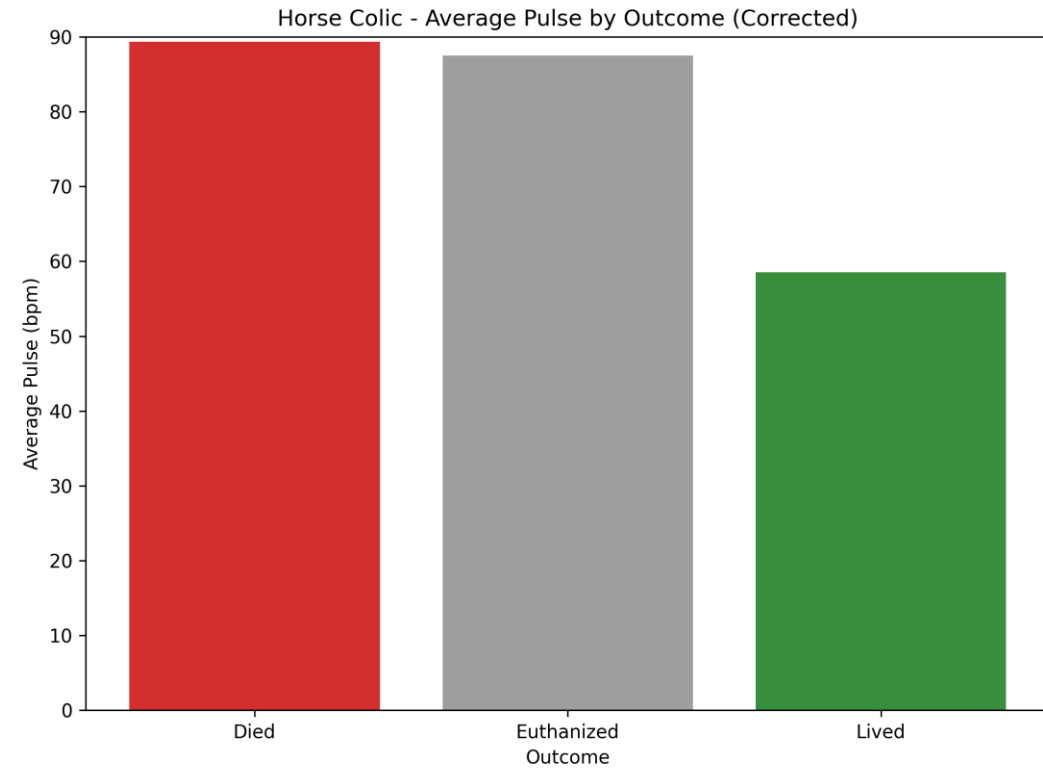
Tomáš Války

Pair 1: Truncated Y-Axis Bar Chart

Misleading Visualization



Corrected Visualization

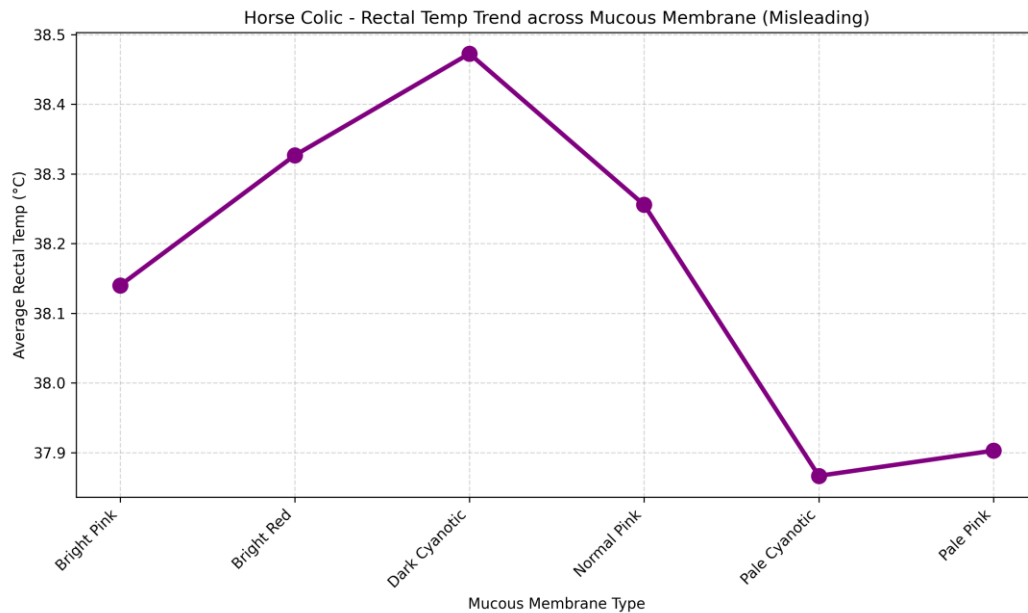


Explanation:

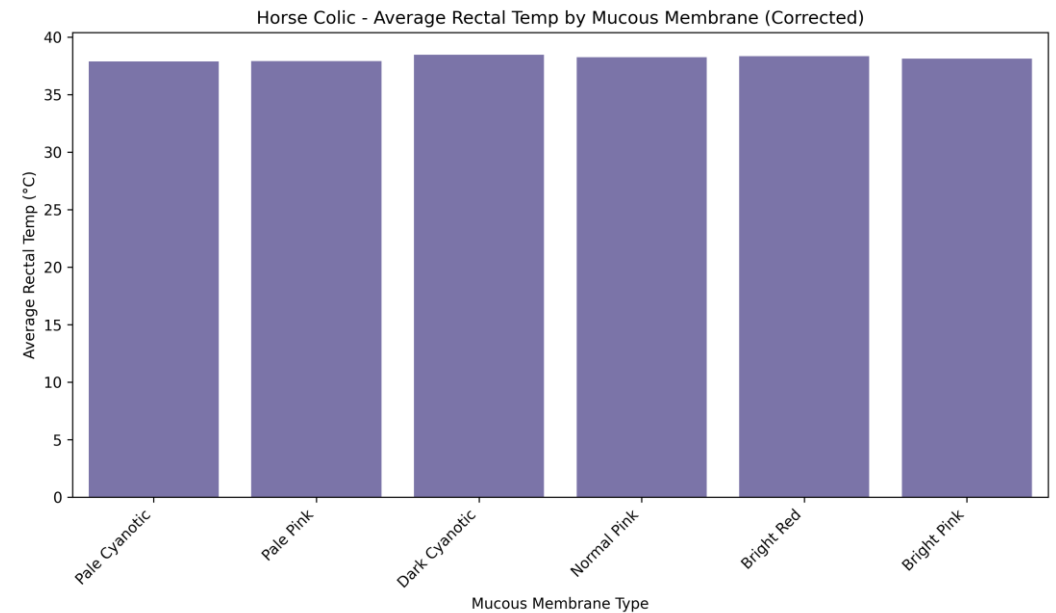
- The misleading visualization (Left) starts the Y-axis at 50 instead of 0. This artificially exaggerates the small difference in average pulse between lived, died, and euthanized horses, making the lower value look almost non-existent.
- The corrected version (Right) fixes this by establishing a zero-baseline, truthfully reflecting that the differences, while present, are not orders of magnitude apart.

Pair 2: Inappropriate Implicit Trend Line

Misleading Visualization



Corrected Visualization

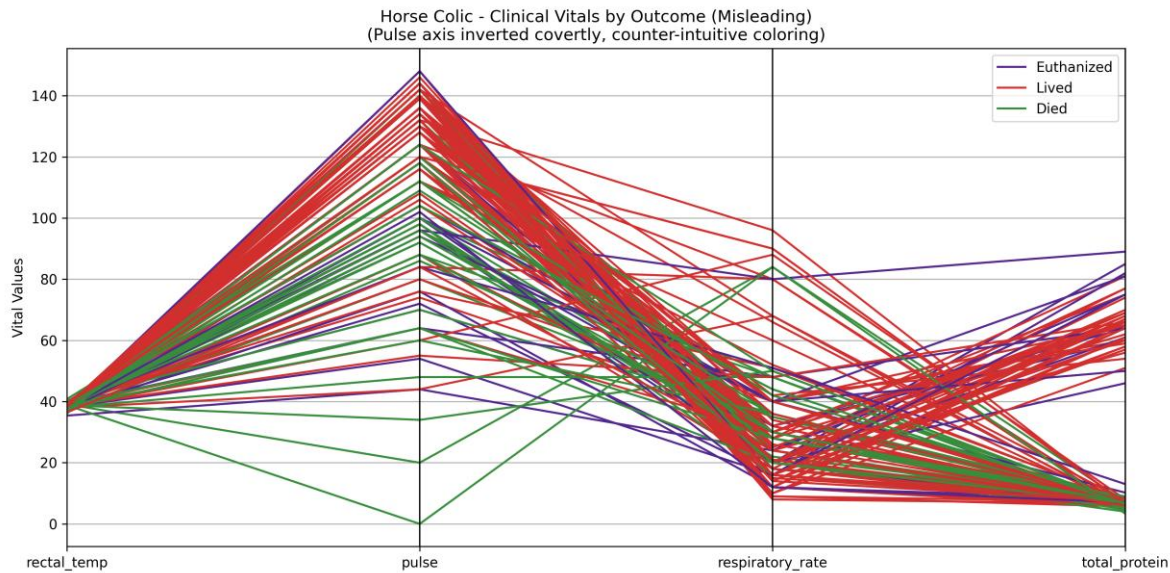


Explanation:

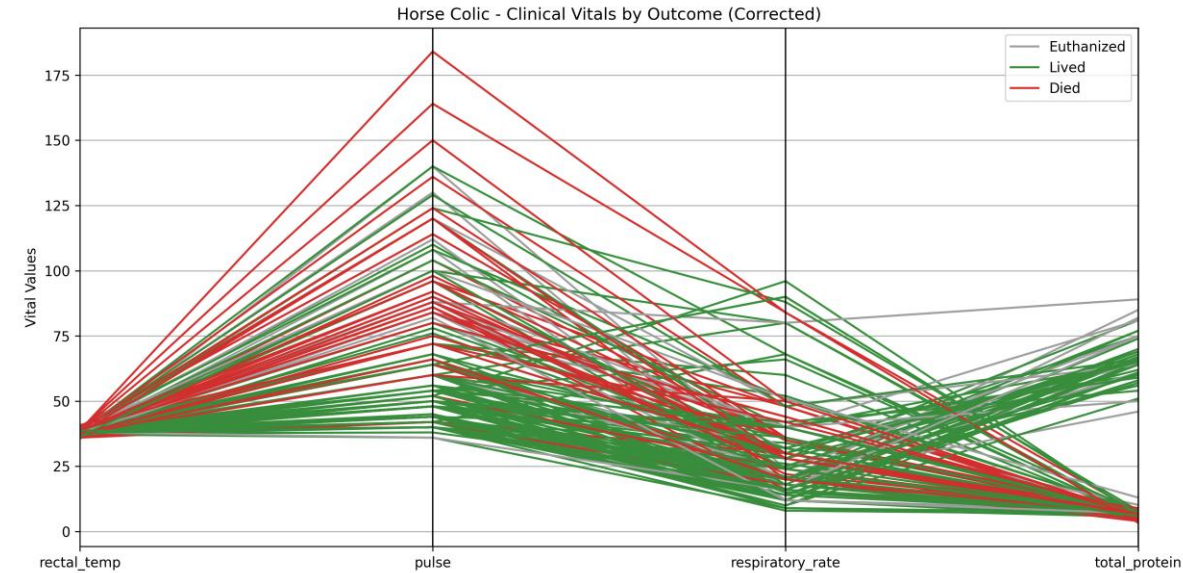
- The misleading visualization (Left) plots categorically distinct 'Mucous Membrane' conditions on the X-axis using a connected line chart. This lies by implying a sequential trend or relationship between discrete conditions.
- The corrected version (Right) replaces the line chart with a Bar Chart, properly handling the categories and removing the false implication of continuity.

Pair 3: Multidimensional (Parallel Coordinates)

Misleading Visualization



Corrected Visualization



Explanation:

- The misleading visualization (Left) flips the 'Pulse' axis manually, making high values plot at the bottom, and assigns 'Lived' to Red and 'Died' to Green (counter-intuitive). This hides the relationship that higher pulses often link to death.
- The corrected version (Right) orients all axes uniformly in the standard direction and uses intuitive semantic colors.

TOOLS, TECHNOLOGIES & AI SUPPORT

- **Programming Languages & Tools**

- Python was selected for automated, reproducible plot generation and document creation.
- Libraries used: pandas (data loading/cleaning), matplotlib & seaborn (generating misleading and corrected visualizations), python-pptx (generating this presentation structure automatically).

- **AI Support**

- DeepMind Agent AI was utilized to:
 - Identify logical flaws to 'lie' effectively with data.
 - Generate the Python code required to produce exactly formatted matplotlib images and construct the presentation output.